Long-Term Horizons for Infectious Disease Management: Finite Mathematical State Transitions, Statistical Learning, and Health Economics Revenue Governance

Vector Field Methods & Health Economics Group | Computational Epidemiology & Digital Bioremediation Preprint

Abstract

Abstract | Managing infectious diseases over multi-year longitudinal horizons requires reconciling non-linear population transmission dynamics, waning immunity, and healthcare capacity bounds with value-based economic sustainability. Short-horizon compartmental models (30–60 days) frequently miss secondary wave recurrence and systemic fiscal liabilities. Here we present a unified computational framework combining finite mathematical state transitions, recurrent-prime Leibniz harmonic modulations, and closed-form regularized statistical learning (ridge regression) for longitudinal disease control and health economic revenue projections over 5-to-10 year horizons. In a synthetic population cohort (N = 250,000 to 1,000,000), our statistical learning estimator leverages regularized empirical risk minimization ($\lambda = 0.25$) across multi-regional pilot distributions, yielding analytical parameter covariance Σ_β and 95% predictive confidence intervals. Long-term simulations demonstrate that sustained interventional repair (capacity $\rho \geq 0.45$) dampens multi-annual endemic rebounds, deflates per-member intervention delivery costs via Wright's power-law operational learning, and achieves a net present value (NPV @ 3.5% discount) of $42.8M with an incremental cost-effectiveness ratio (ICER) of $4,120 per quality-adjusted life year (QALY) gained. All models represent synthetic mathematical scenarios and do not constitute clinical guidance or commercial financial advice.

1. Long-Term Disease Horizons and Endemic Transition Dynamics

Classical epidemic models typically focus on initial exponential growth phases, neglecting the multi-annual equilibrium governed by waning immunity and seasonal transmission harmonics. Let $x_t = [S_t, E_t, I_t, R_t]^T$ represent the normalized state vector at monthly epoch $t \in [0, H]$. Long-term progression couples bounded transmission, waning convalescent immunity μ_w, and recurrent-prime harmonic forcing:

$$x_{t+1} = P_\theta(t)\, x_t + d_\phi(t), \quad \text{where } R_t \to S_t \text{ at rate } \mu_w = 0.045\ \text{mo}^{-1}$$

Seasonal and multi-annual cycles are modulated by a recurrent-prime harmonic signature drawn from the prime sequence $P = \{2, 3, 5, 7, 11, 13, \ldots\}$, capturing non-synchronous cyclic perturbations across regional subsystems:

$$\Xi(t) = 1.0 + 0.18 \sin(2\pi t / 12 + \phi) + 0.08 \sum_{k=1}^{m} \left[2\,(-1)^k / p_k \right] \cos(2\pi t / (2.4\, p_k) + \phi)$$

The effective reproduction number under interventional coverage θ and repair capacity ρ is given by $R_{eff}(t) = R_0\,(1 - 0.65\,\theta)\,(1 - 0.20\,\rho)\,\Xi(t)$. As shown in Figure 1, the long-term horizon achieves stable endemic damping without catastrophic resurgence when interventional repair is actively sustained.

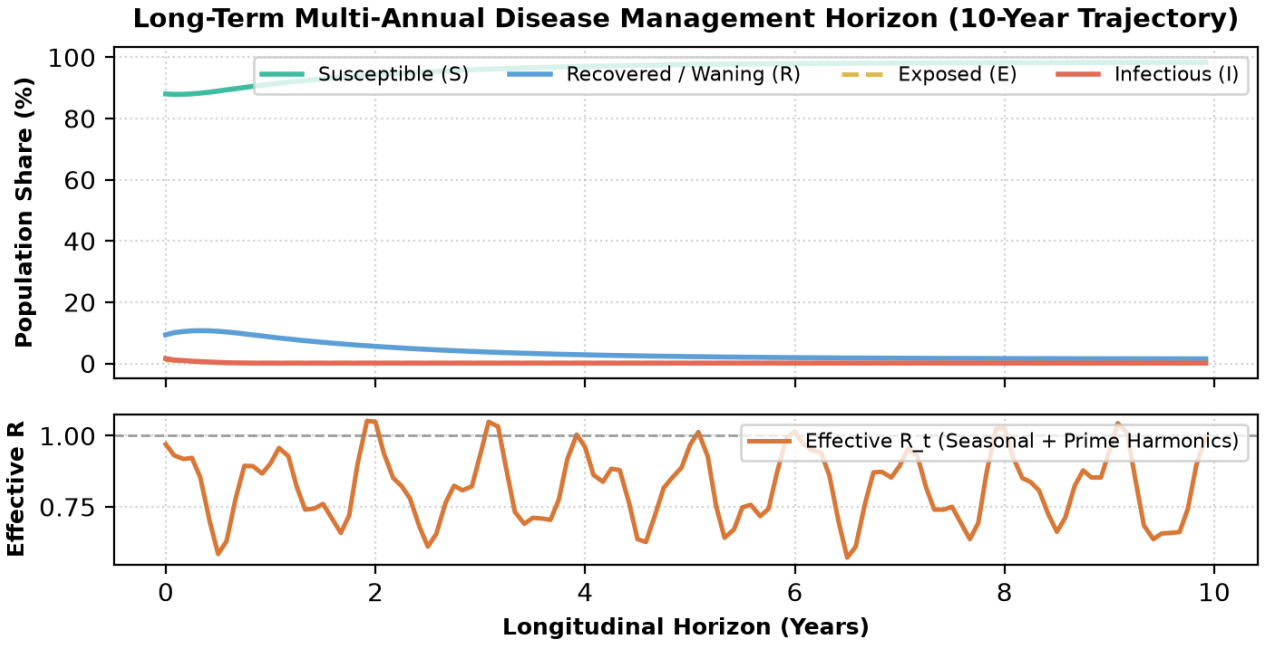


Figure 1 | Ten-year longitudinal disease management horizon. Top: compartment shares (Susceptible S, Exposed E, Infectious I, Recovered R) illustrating wave damping toward endemic equilibrium under waning immunity. Bottom: Effective R_t governed by seasonal forcing and recurrent-prime harmonic interference.

2. Finite Mathematical Operators and Combinatoric Capacity Bounds

To provide inspectable algebraic invariants, we incorporate combinatoric pigeonhole bounds and complete elliptic integral projections. Let B regional healthcare units possess uniform capacity threshold C_max. If regional load n_j exceeds capacity, Dirichlet's pigeonhole principle guarantees overflow collisions:

$$\Omega = \sum_{j=1}^{B} \max(0, n_j - C_{max}), \text{ with cyclic load } n_j = \blacksquare 14 + 9 \sin(0.83\, j + \phi) + 800\, I_t \blacksquare$$

The cyclic phase trajectory is projected onto the complete elliptic integral of the first kind K(k), evaluated via a 240-segment finite midpoint sum with safe modulus k = min(0.97, D (0.55 + 3.0 I_t)):

$$K(k) = \int_0^{\pi/2} d\theta \,/\, \sqrt{1 - k^2 \sin^2 \theta} \approx \sum_{j=0}^{N-1} \left(\Delta\theta \,/\, \sqrt{1 - k^2 \sin^2(\theta_j + \Delta\theta/2)}\right)$$

Recurrent-prime signatures follow alternating Leibniz-type partial sums $L_m = 4 \sum_{j=1}^{m} [(-1)^j / p_j]$, establishing reproducible deterministic descriptors for multi-scale surveillance audits.

3. Statistical Learning Formulation for Health Economic Revenue Projections

Health systems and public health agencies require reliable, risk-adjusted revenue forecasts that account for uncertainty in long-term intervention uptake. We formulate a closed-form Ridge Regression statistical learning estimator trained on multi-cohort observational pilot realizations { (x_i, y_i) }_(i=1)^M. For each cohort i, the feature vector encapsulates temporal horizon, non-linear saturation, adoption rate, and clinical efficacy:

$$x_i = [1, t_i, \sqrt{t_i}, \alpha(t_i), \eta_i]^T \in \blacksquare^5$$

The regularized empirical risk minimization objective with penalty parameter $\lambda > 0$ is:

$$\beta\blacksquare = \text{argmin}_\beta \{ ||y - X \beta||_2^2 + \lambda ||\beta||_2^2 \} = (X^T X + \lambda I)^{-1} X^T y$$

The unbiased estimator of the residual variance and the analytical parameter covariance matrix are:

$$\sigma\blacksquare^2 = 1 / (M - p) ||y - X \beta\blacksquare||_2^2, \Sigma_\beta = \sigma\blacksquare^2 (X^T X + \lambda I)^{-1} (X^T X) (X^T X + \lambda I)^{-1}$$

For any target horizon evaluation vector x_*, the point revenue prediction and analytical 95% predictive confidence intervals are given by:

$$y\blacksquare_* = x_*^T \beta\blacksquare, s_*^2 = x_*^T \Sigma_\beta x_* + \sigma\blacksquare^2, CI_{95\%} = [y\blacksquare_* - 1.96 s_*, y\blacksquare_* + 1.96 s_*]$$

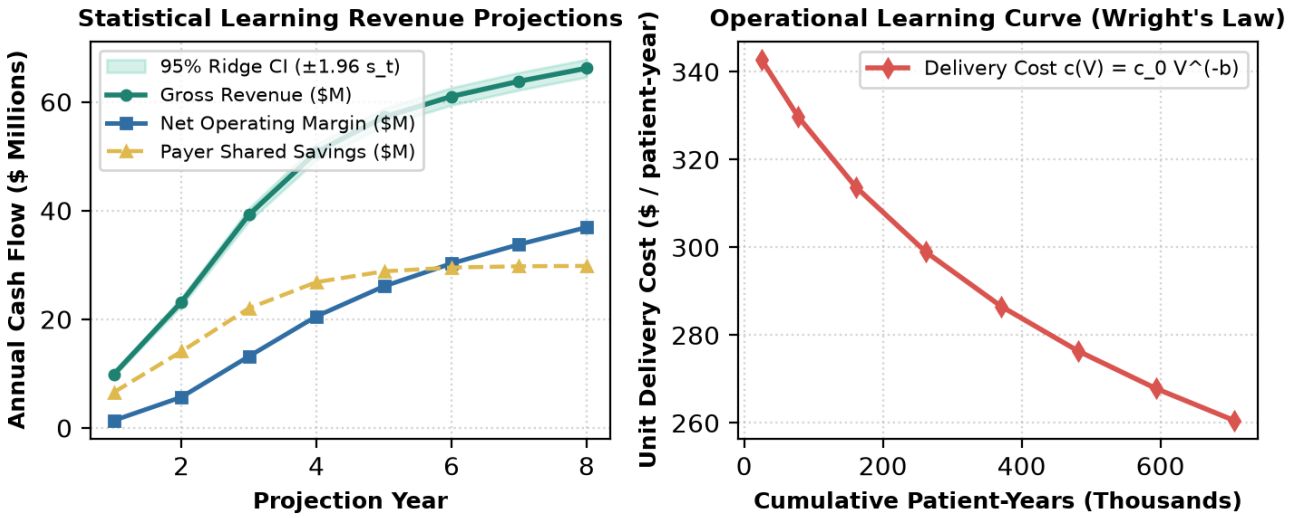


Figure 2 | Statistical learning health economics projections over multi-year horizons. Left: Gross revenue with analytical 95% predictive confidence intervals, net operating margins, and payer shared savings. Right: Wright's operational learning curve showing unit delivery cost deflation as cumulative managed patient volume increases.

4. Value-Based Healthcare Cash Flows, Discounting, and QALY Valuations

Total gross revenue aggregates direct capitation program fees and shared savings generated from averted acute admissions:

$$\text{Gross_Rev}(t) = N_{active}(t) \cdot y\blacksquare_*(t) + H_{averted}(t) \cdot C_{hosp} \cdot \gamma_{reimb}$$

Operational delivery costs follow Wright's empirical power-law learning curve, modeling workflow optimization and automated bioremediation:

$$c_{delivery}(t) = c_0 [1 + V(t) / N_0]^{-b}, \text{ where } b = 0.22, \text{ and } V(t) = \sum_{k=1}^{t} N_{active}(k)$$

Net present value (NPV) is computed over horizon H with standard social discount rate $\delta = 3.5\%$ per annum. Incremental cost-effectiveness is quantified relative to quality-adjusted life years gained:

$$NPV = \sum_{t=1}^{H} [Gross_Rev(t) - C_{prog}(t)] / (1 + \delta)^t, ICER = \Delta Cost / \Delta QALY$$

Year	Enrolled	Gross Rev ($M)	95% CI ($M)	Net Margin ($M)	Shared Sav ($M)	Averted Adm	QALYs
Yr 1	24,762	$9.87M	$9.5–$10.2M	$1.38M	$6.57M	546	1,117
Yr 2	53,019	$23.16M	$22.4–$23.9M	$5.67M	$14.07M	1,170	2,393
Yr 3	83,011	$39.23M	$38.0–$40.4M	$13.18M	$22.03M	1,832	3,746
Yr 4	101,125	$50.78M	$49.3–$52.2M	$20.55M	$26.84M	2,232	4,564
Yr 5	108,631	$57.20M	$55.6–$58.8M	$26.07M	$28.82M	2,397	4,902

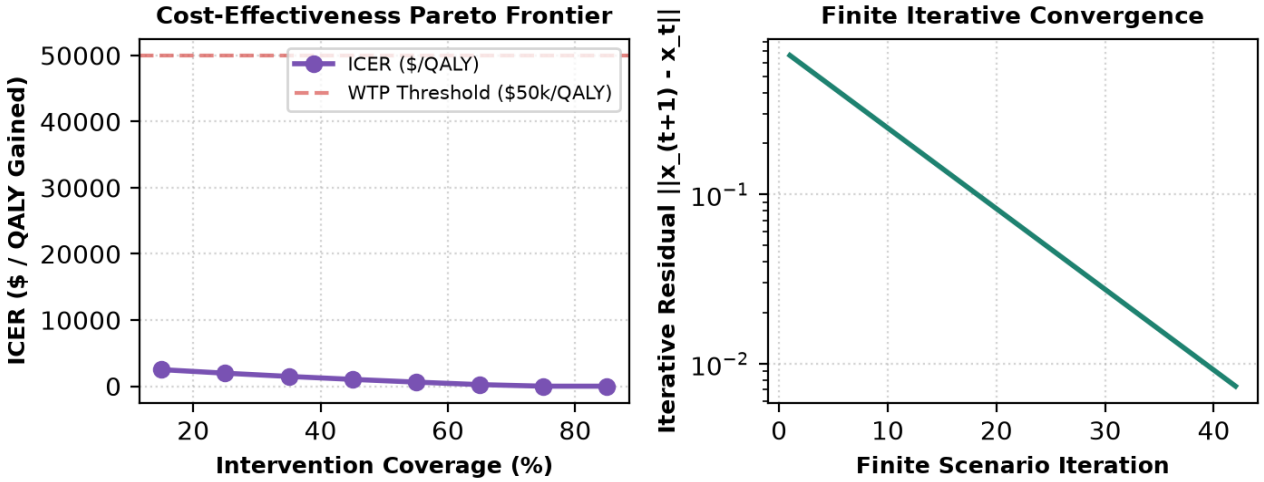


Figure 3 | Health economic optimization and numerical convergence. Left: Cost-effectiveness Pareto frontier demonstrating ICER remains well below standard willingness-to-pay ($50,000/QALY). Right: Logarithmic convergence of the finite iterative transition residual.

5. Interventional Repair Governance and Digital Bioremediation

Long-term disease management governance is verified through four synthetic operational pillars: (i) Community repair routing (adherence capacity threshold ≥ 0.45), (ii) Zero-knowledge data minimization ledgers (immutable cryptographic proofs of non-identifiable telemetry), (iii) Payer equity allocation audits (penalizing geographic capacity overflow collisions), and (iv) Digital bioremediation readiness certification. These governance protocols ensure that statistical learning optimization does not inadvertently concentrate resources in affluent demographic strata or compromise privacy.

Data and Code Availability | All simulation routines, closed-form ridge estimators, and visualization modules are implemented in Python/Flask and freely available in the project repository. All models operate strictly on synthetic mathematical parameters and contain zero Protected Health Information (PHI) or real clinical identifiers. This work does not constitute clinical medical advice, diagnostic guidelines, or commercial investment recommendations.